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Multilevel models for longitudinal data

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Maria Levina, Rahul Vishwkarma

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Outline

Longitudinal data

Models for repeated measures

Multilevel models of growing flexibility

Example: Analysing growth trajectories of children

Recap and outlook

Longitudinal data

Origins of longitudinal data

Longitudinal data: repeated observations of the same research objects (units) over a time period. A unit can be a physical object or a single person as well as a family/household or an organization.

$\Rightarrow$ Both within-person and between-person analysis possible; generalizes over population.

Example: mathematics skills of 10 students during the education period measured by regular tests.

Difference to time series:

Time series is a collection of sequential observations of a single object or entity over time.

$\Rightarrow$ Only within-person/within-entity analysis possible; generalizes over time period.

Example: mathematics skills of high-school students over the last 5 years.

Features of longitudinal data

- Observations of different unit are independent, but they are not independent within unit.
- Different levels of analysis are accessible:
  1. Between-person / inter-individual / level 2 analysis: relationships among inter-individual differences in time-invariant variables.
  2. Within-person / intra-individual / level 1 analysis: relationships among intra-individual differences in time-variant variables.
- Data continuum: analysis of change (systematic increase or decrease over time) in long-term vs. analysis of fluctuation (undirected variation over time) in short-term.

Models for repeated measures

Models for means and variances

Following models have two parts responsible for different aspects of the data combined:

Model for the means: how the expected (or predicted) outcome for each person varies as a function of his or her predictor values.

Predictor weights are called here *fixed effects* since they are constant for everyone in the sample.

Models for means and variances

Following models have two parts responsible for different aspects of the data combined:

Model for the means: how the expected (or predicted) outcome for each person varies as a function of his or her predictor values.

Predictor weights are called here *fixed effects* since they are constant for everyone in the sample.

Model for variance: how the residuals of the outcome are distributed and related across observations. That is, it predicts patterns of variance and covariance in residuals.

Addition of *random effects* allow greater flexibility - variance between and within a person can be distinguished. Random effects allow every unit to have its own effect of the same predictor.

RM-Anova

Traditional method for handling repeated (dependent) measurements. It reproduces the observed mean per occasion using fixed effects equal to the number of occasions minus 1.

Assumptions:

- Normal distribution of residuals;
- Homoscedasticity - constant variance over time points;
- Compound symmetry - equal variances and equal covariances over subjects, or
- Sphericity - equal variance and equal covariance of the pairwise differences between occasions.

Drawbacks:

1. Rather strict assumptions for real-world data;
2. Cannot handle missing data;
3. Requires equal intervals between measurement points;
4. Only two levels of analysis can be accommodated.

Multilevel Models

Assumptions:

- Observations from different subjects are independent, but are correlated within the subject.
- Dependent variable is normally distributed with covariance matrix Σ.

Key strengths:

- Handles missing or unequal timing of observations naturally;
- Works for unbalanced datasets (different numbers of observations per person),
- Can include additional (time-invariant) predictors;
- Captures complex covariance patterns.

Multilevel / Mixed effects model

Mixed models: both fixed and random effects are allowed, i.e. model parameters for covariates are allowed to vary from individual to individual.

Multilevel models: Parameters and their (co-)variances are modelled separately, i.e. on different levels.

Level 1 – Within-person level

- Represents repeated measurements over time for each unit.
- Captures *within-unit variation* (how a single subject characteristic changes with time).

Level 2 – Between-person level

- Represents individuals as units.
- Captures *between-unit variation* (overall differences between subject).

Multilevel models: Variance Matrices

Key idea: *Observed variability comes from both between-person differences and within-person fluctuations.*

- $G \in \mathbb{R}^{u \times u}$: Variances and covariances of $u \in \mathbb{N}$ random effects (between-person variability)
- $R \in \mathbb{R}^{n \times n}$: Residual variances and covariances across $n \in \mathbb{N}$ time points (within-person variability).

Total variance: $V = Z^T G Z + R$, where $Z \in \mathbb{R}^{u \times n}$ is a constructed transformation matrix.

Multilevel models of growing flexibility

Random intercept empty model

The simplest possible model without any effect of covariates.

$$\textit{Level 1: } y_{ti} = \beta_{0i} + e_{ti}, \qquad e_{ti} \sim \mathcal{N}(0, \sigma_e^2)$$
$$\textit{Level 2: } \beta_{0i} = \gamma_{00} + U_{0i}, \qquad U_{i0} \sim \mathcal{N}(0, \tau_{U_0}^2)$$

y_{ti} - observation of dependant variable of unit $i = 1, \ldots, M$, at a time point $t \in t_1, \ldots, t_N$;
γ_{00} - fixed intercept;
U_{0i} - random intercept, varies over individuals;

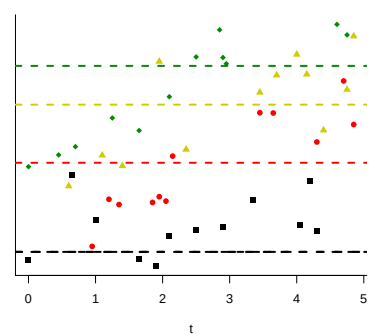


Figure 1: Random intercept empty model

Random intercept, fixed slope model

Simple unconditional model: time is the only predictor.

Level 1: $y_{ti} = \beta_{0i} + \beta_{1i} \cdot \text{Time}_{ti} + e_{ti}, \quad e_{ti} \sim \mathcal{N}(0, \sigma_e^2)$

Level 2:

$\beta_{0i} = \gamma_{00} + U_{0i}$

$\beta_{1i} = \gamma_{10}$

γ_{00} - fixed intercept;

$U_{0i} \sim \mathcal{N}(0, \tau_{U_0}^2)$ - random intercept, varies over individuals;

γ_{10} - fixed slope - linear effect of time;

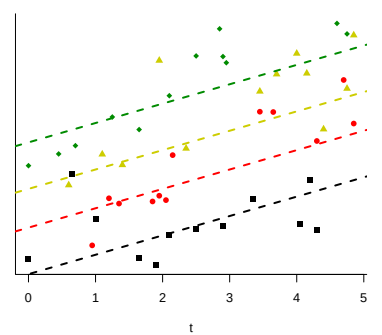


Figure 2: Random intercept, fixed linear time model

Random intercept, random slope model

Time is the only predictor, but it is included as a random effect.

$$\textit{Level 1: } y_{ti} = \beta_{0i} + \beta_{1i} \cdot \text{Time}_{ti} + e_{ti}, \quad e_{ti} \sim \mathcal{N}(0, \sigma_e^2)$$

Level 2:

$$\beta_{0i} = \gamma_{00} + U_{0i}, \quad U_{0i} \sim \mathcal{N}(0, \tau_{U_0}^2)$$
$$\beta_{1i} = \gamma_{10} + U_{1i} \quad U_{1i} \sim \mathcal{N}(0, \tau_{U_1}^2)$$

γ_{00}, γ_{10} - fixed parameters;
U_{0i}, U_{1i} - random parameters;
In vector notation for individual i:

$$Y_i = X_i \cdot \gamma + Z_i \cdot U_i + E_i$$

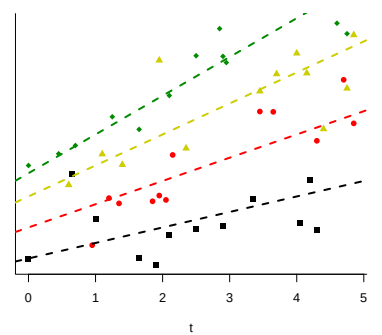


Figure 3: Random linear time model

Random linear time model: Model of variance

Using modelling equation we get: $\mathbb{V}ar(y_{ti}) = \mathbb{V}ar(\beta_{0i} + \beta_{1i} \cdot \text{Time}_{ti} + e_{ti}) =$

$\mathbb{V}ar(\gamma_{00} + U_{0i}) + \text{Time}_{ti}^2 \cdot \mathbb{V}ar(\gamma_{10} + U_{1i}) + \mathbb{V}ar(e_{ti}) = \tau_{U_0}^2 + \text{Time}_{ti}^2 \cdot \tau_{U_1}^2 + \sigma_e^2$

Or in matrix notation: $V_i = \mathbb{V}ar(Y_i) = \mathbb{V}ar(X_i \cdot \gamma + Z_i \cdot U_i + E_i) = Z_i^T \mathbb{V}ar(U_i) Z_i + R = Z_i^T G Z_i + R.$

$$\text{Here:} \qquad X_i = \begin{pmatrix} 1 & t_1 \\ \vdots & \vdots \\ 1 & t_{n_i} \end{pmatrix} \in \mathbb{R}^{n_i \times f} \quad Z_i = \begin{pmatrix} 1 & t_1 \\ \vdots & \vdots \\ 1 & t_{n_i} \end{pmatrix} \in \mathbb{R}^{n_i \times u},$$

where f - number of fixed effects, and u - number of random effects in the model.

Due to the construction of the Z matrix model can accommodate data from irregular observations with different number of time n_i points per person i.

Model assumptions

1. $E_i \sim \mathcal{N}(0, R)$, $R = \sigma_e^2 I_{n_i}$, however, another covariance structures (such as AR(1)) can be assumed (restricted likelihood test (nested structures), and AIC or BIC (not nested)).
2. $U_i \sim \mathcal{N}(0, G)$, $G = \begin{pmatrix} \tau_{U_0}^2 & \tau_{U_{01}} \\ \tau_{U_{01}} & \tau_{U_1}^2 \end{pmatrix}$ - unstructured covariance matrix.
3. No covariances are allowed for the variance terms across levels.
4. All variances and covariances are assumed constant over persons.
5. All units are assumed conditionally independent $\Leftrightarrow$ no relationships within each of the U_{0i}, U_{1i}, or e_{ti} values across persons after including the model predictors.

Conditional multilevel models

Additional predictors: time-invariant characteristics; Baseline observation (at the beginning of the study) or constant features.

Example: blood pressure before treatment, gender, personal traits etc.

Level 1: $y_{ti} = (\gamma_{00} + U_{0i}) + (\gamma_{10} + U_{1i}) \cdot \text{Time}_{ti} + \gamma_{20} \cdot \text{Feature}_{1i} + \ldots + \gamma_{k+1\,0} \cdot \text{Feature}_{ki} + e_{ti}$

Level 2: unchanged

$\Rightarrow Z_i \in \mathbb{R}^{n_i \times 2}$ and $X_i \in \mathbb{R}^{n_i \times (2+k)}$, where k - number of add. covariates in the model.

Optional conceptual extension:

- Cross-level interactions: person-level traits predicting the rate of change
- Example: baseline weight moderating weight loss trajectory

Beyond Linear Change

Other types of growth models:

- Polynomial growth models (quadratic, cubic) for accelerating or decelerating trends;
- Non-linear trajectories capturing more complex patterns over time;
- Piecewise/discontinuous models for interventions or events.

Example: add a quadratic term of time into random linear model
$\Rightarrow$ additional column in Z_i and $G \in \mathbb{R}^{3\times 3}$.

Parameter estimation: Maximum likelihood

Maximum-likelihood function to optimize can be written as follows:

$$L = \prod_{i=1}^{N} (2\pi)^{-\frac{n}{2}} \cdot |V_i|^{-\frac{1}{2}} \cdot \exp\left[-\frac{1}{2}(Y_i - X_i\gamma)^T V_i^{-1}(Y_i - X_i\gamma)\right]$$

with log-likelihood:

$$LL = -\frac{T}{2}\log(2\pi) - \frac{1}{2}\sum_{i=1}^{N}\log|V_i| - \frac{1}{2}\sum_{i=1}^{N}(Y_i - X_i\gamma)^T V_i^{-1}(Y_i - X_i\gamma)$$

Random effects U_i are not included in the likelihood $\rightarrow$ U_i can be predicted with V_i, X_i and Y_i given: $U_i = G_i Z_i^T V_i^{-1}(Y_i - X_i\gamma)$.

Fixed effects γ and their SEs also can be estimated with V_i's given (similar to GLM estimation):

$$\gamma = \left(\sum_{i=1}^{N}(X_i^T V_i^{-1} X_i)\right)^{-1} \sum_{i=1}^{N}(X_i^T V_i^{-1} Y_i) \quad \text{and} \quad \mathbb{C}ov(\gamma) = \left(\sum_{i=1}^{N}(X_i^T V_i^{-1} X_i)\right)^{-1}$$

Parameter estimation: Maximum likelihood

Newton-Raphson algorithm is then used update estimates of matrix V_i iteratively using derivatives of LL-function.

R package `lme4` produces estimates of model parameters using restricted or unrestricted maximum-likelihood method.

Restricted ML: accounts for uncertainty from simultaneous estimation of fixed parameters $\rightarrow$ unbiased parameter variance estimates.

$$LL_{\text{REML}} = -\frac{T-f}{2}\log(2\pi) - \frac{1}{2}\sum_{i=1}^{N}\log|V_i| - \frac{1}{2}\sum_{i=1}^{N}(Y_i - X_i\gamma)^T V_i^{-1}(Y_i - X_i\gamma) - \frac{1}{2}\sum_{i=1}^{N}\log|X_i^T V_i^{-1} X_i|$$

Criteria for model comparison

Many different model configurations possible $\rightarrow$ need for model comparison.

For random effects:

- Two nested models: Likelihood ratio test
  $LR = 2\log\left({}^{L}\text{fewer}/{}^{L}\text{more}\right) = -2(LL_{\text{fewer}} - LL\text{more}) \sim \chi^2_d$, where d is difference in number of model parameters;
  REML results can only be used to compare two models with the same fixed effects.
- Two non-nested models: $AIC = -2LL + 2p$ and $BIC = 2LL + log(N) \cdot p$. The smaller value of AIC or BIC indicates a better model.

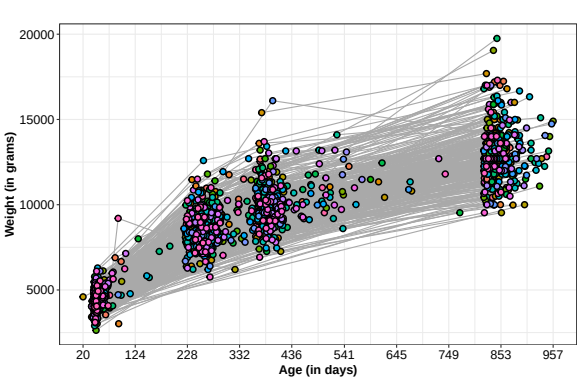
For fixed effects: Wald test for single effect using $\frac{\gamma_i}{SE(\gamma_i)} \sim t_{N-f}$, where f - number of fixed parameters.

Example: Analysing growth trajectories of children

Dataset Overview

Asian Children's Weight Measurements

- Repeated weight measurements of 568 Asian children living in a British community.
- Measures at ≈ target examination dates at 6 weeks, 8 months, 12 months, and 27 months.
- Age in days, weight in grams.
- Additional variables:
  gender (1 = boy, 2 = girl),
  birth weight (in grams).



Children's weight observations over time

Dataset Overview: (Im-)balance in the covariates

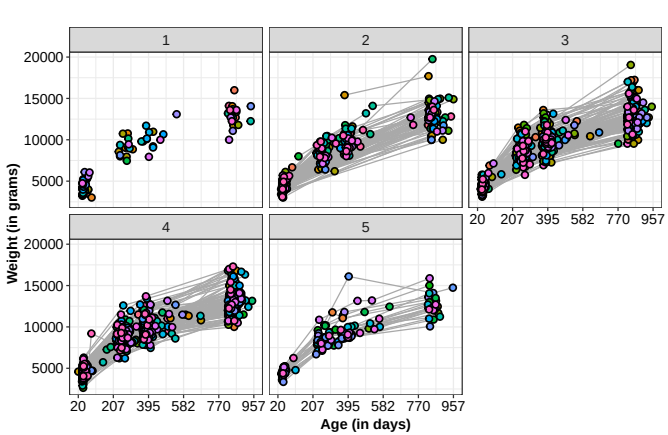


Figure 4: Original observations by number of visits in the clinic

Table 1: Distribution of number of observations

# of visits	Boy	Girl
1	45	46
2	62	79
3	86	83
4	75	68
5	15	9
Total	283	285

Analysis perspective

Research questions:

1. How children's weight changes in the early stages of life?
2. Whether and how those growth patterns differ between individuals?
3. Are there any differences between children of different gender?
4. Is there any influence of birth weight on post-natal development?

Data features:

1. Repeated measurements $\rightarrow$ observations within person are not independent.
2. Individual differences in growth $\rightarrow$ Children start at different baseline weights and might grow at different rates.
3. Unequal number of observations & irregular timing $\rightarrow$ Some children have fewer visits or varying ages at measurement

Why multilevel models? Measurements are nested within children — a model that captures both levels of variation is required.

Data preprocessing

Different strategies to ensure numerical stability and to avoid influences of different scales on the model estimation:

- Rescaling: days into weeks or years, grams into kilograms;
- Centralization (subtracting general mean over time and individuals);
- Logarithm of the values;
- Normalization (subtracting general mean over time and individuals and dividing by standard deviation).

For further analysis we used centred age in years and centred weight in kilograms. We removed all units with only 1 observation for convergence.

Example: Unconditional linear models

We tested 3 different model configurations.

Table 2: Multilevel linear time model specifications

ID	β_0	β_{Time}	β_{Time^2}	Birthweight	Gender	Model	*AIC*	$-2\Delta LL$	*p*-value
01	random	-	-	-	-	singular	7708.2	7702.2	
02	random	fixed	-	-	-	+	5357.5	5349.5	$< 2.2e-16$
03	random	random	-	-	-	singular	5286.2	5274.2	$< 2.2e-16$

Table 3: Fixed effects estimates for the linear growth model

Effect	Estimate	Std. Error	df	*t* value	*p*-value
(Intercept)	0.0106	0.0485	456.8	0.219	0.827
Age	3.6750	0.0449	1139.0	81.746	$< 2 \times 10^{-16}$

Random intercept variance: $\hat{\tau}_{U_0} = 0.5329 \pm 0.730$. Residual variance $\hat{\sigma}_e = 1.7598 \pm 1.327$

Example: Conditional random intercept, fixed slope model

Then we added time-invariant covariates: Birthweight (also centered, in kg) and Gender (binary encoded, 0 = Boy, 1 = Girl) - as well as interactions.

$$\text{Weight}_{it} = \beta_0 + U_{0i} + \beta_1 \text{Age}_{it} + \beta_2 \text{Birthweight}_i + \beta_3 \text{Gender}_i + \beta_4 \text{Age}_i \cdot \text{Birthweight}_i + \beta_5 \text{Age}_i \cdot \text{Gender}_i,$$

Table 4: Fixed effects estimates from the multilevel model

Predictor	Estimate	Std. Error	df	t value	p-value
(Intercept)	0.26149	0.05842	429.17	4.48	9.77×10^{-6}***
Age	3.71426	0.06265	1168.81	59.28	$< 2 \times 10^{-16}$***
Birthweight	0.91021	0.08326	430.57	10.93	$< 2 \times 10^{-16}$***
GenderID	-0.49726	0.08330	442.90	-5.97	4.88×10^{-9}***
Age × Birthweight	0.18107	0.08929	1183.34	2.03	0.0428*
Age × GenderID	-0.09151	0.08915	1172.61	-1.03	0.3049

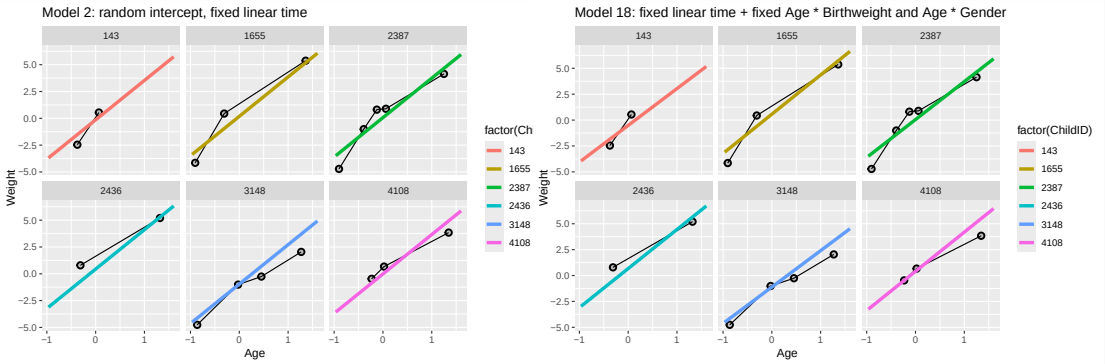
Note:

***$p < 0.001$, **$p < 0.01$, *$p < 0.05$.

Random intercept variance: $\hat{\tau}_{U_0} = 0.2449 \pm 0.4949$. Residual variance $\hat{\sigma}_e = 1.7591 \pm 1.3263$.

Example: Unconditional vs. conditional RIFS model

ID	β_0	β_{Time}	Birthweight	Gender	Model	AIC	$-2\Delta LL$	p-value
01	random	-	-	-	singular	7708.2	7702.2	
02	random	fixed	-	-	+	5357.5	5349.5	$< 2.2e-16$
03	random	random	-	-	singular	5286.2	5274.2	$< 2.2e-16$
18	random	fixed	interaction	interaction	+	5219.0	5203.0	$3.412e-16$



Example: Modelling quadratic time

Table 5: Multilevel quadratic time unconditional models

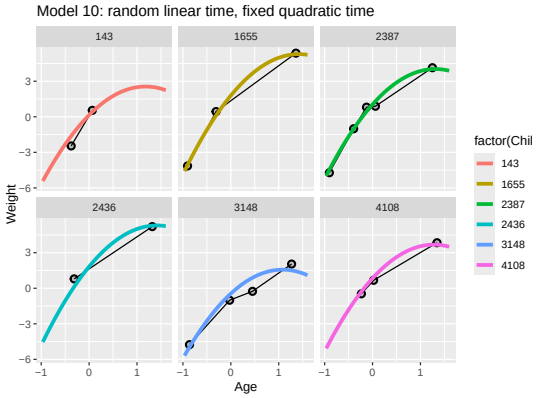
ID	Intercept	Time	Time^2	Birthweight	Gender	Model	AIC	$-2\Delta LL$	p-value
02	random	fixed	-	-	-	+	5357.5	5349.5	$< 2.2e-16$
09	random	fixed	fixed	-	-	+	4356.0	4346.0	$< 2.2e-16$
10	random	random	fixed	-	-	+	4080.5	4066.5	$< 2.2e-16$
11	random	random	random	-	-	singular	3910.7	3890.7	$< 2.2e-16$

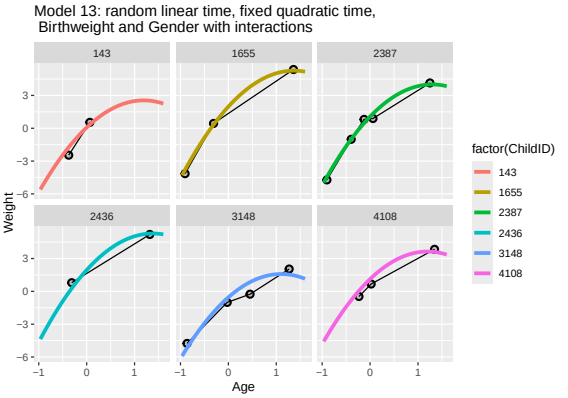
Example: Conditional model for quadratic time

Table 6: Multilevel quadratic time conditional models

ID	Intercept	Time	Time^2	Birthweight	Gender	Model	AIC	$-2\Delta LL$	p-value
09	random	fixed	fixed	-	-	+	4356.0	4346.0	
12	random	fixed	fixed	interaction	interaction	+	4173.7	4155.7	$< 2.2e-16$
10	random	random	fixed	-	-	+	4080.5	4066.5	
13	random	random	fixed	interaction	interaction	+	3847.7	3825.7	$< 2.2e-16$

Example: Unconditional vs Conditional model for quadratic time





Example: Summary

- Data preprocessing is crucial - for convergence and clear interpretation.
- Dataset imbalance is handled quite well by multilevel models.
- Linear time: the best model is random intercept, fixed time model. Improvement possible.
- Quadratic time: both fixed and random linear time effects fit, fixed qaudratic effect makes more sense.
- Outlook: other non-linear transformations (like $log()$) can be used to model growth trajectories.

Recap and outlook

Key Points Recap

Main ideas:

- Longitudinal data involve repeated measures nested within individuals → need multilevel approach
- Multilevel models separate variation into within-person and between-person parts
- Fixed effects capture group trends; random effects capture individual differences in those trends
- These models handle missing data, irregular timing, and complex patterns flexibly.

Concluding statement:

Multilevel modeling provides a powerful framework to understand both general patterns and personal trajectories in longitudinal research.